

# TRACE User Guide

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None

*TRACE Development Team*

*None*

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# 1. Welcome to TRACE

**TRACE (Triggered Radiotherapy Adaptive Control Engine)** is a dedicated software suite designed to control and verify dynamic anthropomorphic thorax phantoms used in 4D Radiotherapy (4DRT), motion management research, and image-guided radiation therapy (IGRT).

As described in *Lustermans et al. (2024)*, TRACE interfaces with multi-axis motion hardware and triggering electronics to simulate realistic patient breathing motions and evaluate 4DCT and radiotherapy treatment delivery.

## 1.1 Key Features

- **Analytical & Patient-Derived Motion Profiles:**
  - Generate synthetic respiratory waveforms (sinusoidal,  $\cos^4$ ), custom analytical functions).
  - Import and process patient-derived respiratory curves from 4DCT/RPM clinical logs.
  - Scale, smooth, adjust baseline drift, and resample motion profiles.
- **Multi-Axis Trajectory Planning:**
  - Full 6-DOF support (Translational: SI, AP, LR; Rotational: Roll, Pitch, Yaw).
  - Automated G-code generation optimized for Duet motion controllers.
- **Hardware Integration & Real-Time Control:**
  - Direct communication with Duet 3D motion platform controllers.
  - Real-time axis homing, position tracking, and manual jogging.
- **Triggered Imaging & Verification:**
  - Hardware synchronization with Geiger module for triggered imaging acquisition.
  - Live plotting and position error monitoring during motion execution.

## 1.2 User Guide Overview

This User Guide provides end-to-end instructions for setting up, operating, and troubleshooting TRACE:

1. **Installation:** Setting up the Python environment and launching TRACE.
2. **Hardware Setup:** Connecting the Duet controller, phantom stage, and Geiger trigger module.
3. **Files & Project Management:** Managing project directories and patient motion datasets.
4. **Motion Planning & G-code:** Creating analytical waveforms, editing patient curves, and exporting G-code.
5. **Phantom Control & Execution:** Homing, running motion trajectories, and triggering.
6. **Real-Time Status & Monitoring:** Live verification, logging, and performance metrics.
7. **Troubleshooting & Safety:** Resolving connection issues, limit switch stops, and safety protocols.



### Download PDF Manual

When built with the `with-pdf` plugin, the complete documentation is available as a single downloadable PDF manual under `exports/TRACE_User_Guide.pdf`.

## 2. Installation & Software Setup

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This page guides you through installing the required dependencies and launching the TRACE Graphical User Interface (GUI).

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### 2.1 Prerequisites

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Before installing TRACE, ensure you have installed:

- **Python 3.10 or later** (Conda or Miniconda environment recommended).
  - **Git** (for repository cloning).
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### 2.2 Installation Steps

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#### Step 1: Clone the Repository

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Clone the TRACE repository using Git or your preferred Git client (e.g., SourceTree, GitHub Desktop):

```
git clone https://github.com/YourUsername/TRACE.git
cd TRACE
```

#### Step 2: Create Conda Environment

---

TRACE provides an `environment.yml` file containing all required packages (PyQt / DearPyGui, NumPy, SciPy, Matplotlib, PySerial, etc.).

Create and activate the environment:

```
conda env create -f environment.yml
conda activate trace-env
```

Alternatively, if installing dependencies manually via `pip`:

```
pip install numpy scipy matplotlib pyqt5 pyserial requests
```

---

### 2.3 Configuration ( `configuration.json` )

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Check `configuration.json` in the root directory to verify hardware settings (IP address of Duet controller, default COM ports for Geiger module, default baud rates):

```
{
  "duet_ip": "192.168.1.100",
  "geiger_port": "COM3",
  "baud_rate": 115250,
  "default_output_dir": "./output"
}
```

---

### 2.4 Launching TRACE

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To launch the graphical user interface, run `Launch_ImGUI.py`:

```
python Launch_ImGUI.py
```

 **First Launch Verification**

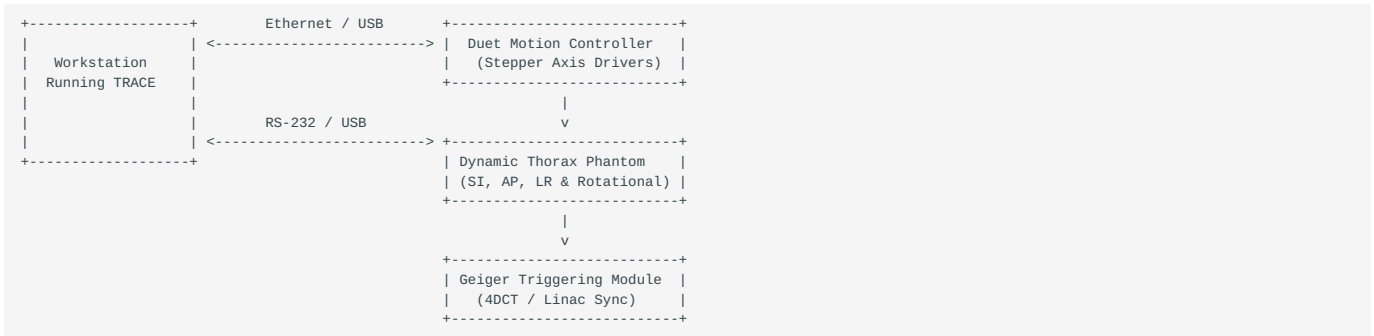
Upon successful launch, the TRACE interface window will open, displaying the navigation tabs (**Files, Planning, Control, Status**) and initializing the communication threads in the background.

## 3. Hardware Setup & Wiring

This guide details the physical hardware setup for connecting the dynamic anthropomorphic thorax phantom, the motion platform controller, and the triggering hardware to TRACE.

### 3.1 Hardware Architecture

The TRACE ecosystem consists of three main hardware components:



#### 3.2 1. Duet Motion Controller Setup

TRACE communicates with a **Duet 3D motion controller** via HTTP REST API or Direct Serial Interface.

1. **Network Cable:** Connect an Ethernet cable from your workstation/switch to the Duet Ethernet port.
2. **IP Configuration:** Ensure the IP address matches `duet_ip` in `configuration.json` (Default: `192.168.1.100`).
3. **Axis Assignments:**
  - **Axis X:** Superior-Inferior (SI) translation.
  - **Axis Y:** Anterior-Posterior (AP) translation.
  - **Axis Z:** Left-Right (LR) translation.
  - **Axis U / V / W:** Rotational axes (Roll, Pitch, Yaw).

#### Emergency Stop Button

Always test the hardware E-STOP button wired to the Duet controller before running automated motion sequences.

#### 3.3 2. Geiger Triggering Module Setup

The Geiger module enables synchronized X-ray imaging / 4DCT triggering during specific phases of phantom motion.

1. **Serial Connection:** Plug the USB-to-Serial adapter from the Geiger module into the workstation.
2. **Identify Port:** Note the assigned COM port (Windows: `COMx`, Linux/macOS: `/dev/ttyUSBx`).
3. **Trigger Output:** Connect the opto-isolated trigger output cable to the external trigger input of the imaging modality (Linac / CT scanner).

### 3.4 3. Power-On Checklist

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Before launching TRACE:

- [ ] Verify 24V power supply connected to Duet controller.
- [ ] Confirm phantom motion stage is free from physical obstacles.
- [ ] Confirm Geiger module LED power indicator is green.
- [ ] Ping Duet IP ( `ping 192.168.1.100` ) from command prompt.



## 4. Chapter 1: Files & Project Management

The **Files Tab** in TRACE ( `tab_files.py` ) is the entry point for importing raw patient motion data, managing analytical waveform presets, and selecting output directories for generated G-code files.

### 4.1 Files Tab Overview

When you open the **Files** tab, you will find:

- **Project Directory Selector:** Choose where motion data and G-code outputs are saved.
- **Raw Data Import Panel:** Load CSV, TSV, DICOM, or log files containing respiratory motion traces.
- **Preset Library:** Select pre-defined standard motion templates (e.g., standard sinusoidal breathing, irregular breathing, high-frequency motion).

### 4.2 1. Importing Patient Respiratory Curves

TRACE supports importing patient-derived motion profiles captured by clinical motion tracking systems (such as RPM, Anzai, or Varian RGSC).

#### Supported File Formats

| Format    | Extensions                                                | Description                                            |
|-----------|-----------------------------------------------------------|--------------------------------------------------------|
| CSV / TSV | <code>.csv</code> , <code>.tsv</code> , <code>.txt</code> | Columnar data containing time vs displacement columns. |
| RPM Logs  | <code>.vms</code> , <code>.txt</code>                     | Varian Real-Time Position Management log files.        |
| G-code    | <code>.gcode</code> , <code>.g</code>                     | Pre-compiled Duet G-code trajectory files.             |

#### Step-by-Step Import Procedure

1. Click **Browse File** under the *Import Motion Data* panel.
2. Select your raw motion file.
3. Specify the time column index and motion axis column index.
4. Click **Parse File**. TRACE will render a preview plot of the imported waveform.

### 4.3 2. Managing Project Directory & Outputs

Outputs generated by TRACE (such as compiled `.gcode` files and trajectory logs) are stored in the selected Project Directory.

- **Set Default Directory:** Use **Select Directory** to choose your target folder.
- **File Naming Convention:**

```
<project_name>_<axis>_<waveform_type>.gcode
```

Example: `planned_motion_platform_SI_sin.gcode`

 **Organizing Patient Datasets**

We recommend creating a separate subfolder for each patient or phantom study inside your project directory to keep G-code and verification logs organized.

## 5. Chapter 2: Motion Planning & G-code Export

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The **Planning Tab** (`tab_planning.py` & `fcn_plan`) is where motion profiles are designed, edited, mathematically transformed, and compiled into controller-ready G-code.

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### 5.1 1. Creating Analytical Waveforms

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TRACE allows you to generate synthetic mathematical motion curves to simulate standard breathing patterns or mechanical test signals.

#### Available Analytical Waveform Types

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1. **Sinusoidal Curve:**  $z(t) = A \cdot \sin\left(\frac{2\pi}{T} t + \phi\right) + z_0$
  2.  $A$ : Amplitude (mm)
  3.  $T$ : Breathing period (seconds)
  4.  $\phi$ : Phase offset (rad)
  5.  $z_0$ : Baseline offset (mm)
  6. **Modified  $\cos^4$  Breathing Curve:** Simulates realistic asymmetry between inspiration and expiration phases:  $z(t) = A \cdot \cos^4\left(\frac{\pi}{T} t\right) + z_0$
  7. **Trapezoidal / Step Motion:** Used for step response tests and static phantom offset calibrations.
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### 5.2 2. Editing Patient-Derived Motion Curves

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Raw clinical respiratory traces often contain baseline drift, high-frequency noise, or cardiac artifacts. TRACE provides interactive tools in `fcn_edit.py` to refine these profiles:

- **Baseline Correction:** Removes gradual baseline elevation drift over time.
  - **Amplitude Scaling:** Rescales motion to match specific phantom limits (e.g., max 30 mm SI displacement).
  - **Smoothing / Filtering:** Applies Savitzky-Golay or low-pass Butterworth filtering to eliminate signal noise.
  - **Resampling:** Interpolates motion data to match the Duet motor update frequency (typically 50 Hz or 100 Hz).
- 

### 5.3 3. Multi-Axis Preview

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Before exporting G-code, preview the 6-DOF motion components in the interactive plot panel:

- **SI Axis (Z-motion):** Primary Superior-Inferior displacement.
  - **AP Axis (Y-motion):** Anterior-Posterior displacement.
  - **LR Axis (X-motion):** Left-Right displacement.
  - **Rotational Axes (Roll, Pitch, Yaw):** Rotational tilting.
-

## 5.4 4. Exporting G-Code for Duet Controller

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Once you are satisfied with the trajectory:

1. Select target motor acceleration and velocity limits.
2. Click **Generate G-Code** ( `fcn_gcode.py` ).
3. TRACE compiles linear/curved segment commands:

```
G90 ; Absolute positioning mode
G1 X12.4 Y2.1 Z-5.0 F1200 ; Linear move with feedrate 1200 mm/min
G1 X12.8 Y2.3 Z-4.8 F1250
```

4. Click **Save G-Code** to store the `.gcode` file into your project output directory.

## 6. Chapter 3: Phantom Control & Execution

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The **Control Tab** (`tab_control.py` & `fcn_control.py`) provides direct control over the dynamic thorax phantom hardware, motor initialization, manual jogging, G-code execution, and trigger synchronization.

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### 6.1 1. Connecting to Hardware

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1. Navigate to the **Control** tab.
  2. Select connection mode: **Ethernet (REST API)** or **Serial (COM)**.
  3. Click **Connect Controller**.
  4. Check connection status indicator:
    - **Connected:** IP / COM port open and responding.
    - **Disconnected:** Verify wiring or IP settings in `configuration.json`.
- 

### 6.2 2. Homing & Zeroing Axes

---

Before executing any motion trajectory, the physical platform **MUST** be homed to establish mechanical zero reference points.

1. Ensure the platform path is clear.
2. Click **Home All Axes** (`\(G28\)`).
3. Wait for all axis status indicators to turn green.
4. Verify position readout shows `(0.00, 0.00, 0.00)`.

#### Axis Safety Limits

Never attempt to run G-code without homing first. Running motion profiles from un-homed positions can cause mechanical collision against endstops.

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### 6.3 3. Manual Jogging

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Use the manual jog panel to fine-tune phantom position or perform pre-experiment align checks:

- **Step Sizes:** `\(0.1\text{mm}, 1.0\text{mm}, 5.0\text{mm}, 10.0\text{mm}\)`.
  - **Directions:** `\(pm X, pm Y, pm Z, pm \text{Roll}, pm \text{Pitch}, pm \text{Yaw}\)`.
  - **Feedrate Control:** Adjustable speed slider (`\(10 - 3000\text{mm/min}\)`).
-

## 6.4 4. Running Motion Trajectories

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To execute a planned G-code motion file:

1. Click **Select G-Code File** and pick your compiled trajectory.
  2. Select execution mode:
    - **Single Cycle:** Plays motion sequence once and returns to home.
    - **Continuous Loop:** Repeats motion sequence until manually stopped.
  3. Click **Start Motion**.
  4. Use **Pause Motion** ( $\backslash(M600\backslash)$ ) or **Emergency Stop** ( $\backslash(M112\backslash)$ ) at any time to halt movement.
- 

## 6.5 5. Geiger Triggering Configuration

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To enable imaging/gating triggers during specific phases of the trajectory:

1. Toggle **Enable Geiger Trigger Sync**.
2. Set **Phase Trigger Point** (e.g., peak inspiration at  $\backslash(90^\circ\backslash)$  or end expiration at  $\backslash(270^\circ\backslash)$ ).
3. Set **Pulse Duration** (e.g.,  $\backslash(50\backslash\text{ms}\backslash)$ ).
4. Click **Test Trigger Signal** to verify optocoupler pulse output on the Geiger hardware module.

## 7. Chapter 4: Real-Time Status & Monitoring

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The **Status Tab** (`tab_status.py`, `fcn_duet.py`, `fcn_plotting.py`) monitors motion execution in real time, displays live feedback plots, and logs trigger events for post-experiment validation.

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### 7.1 1. Live Motion Tracking Plots

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During motion execution, TRACE renders real-time multi-axis tracking plots:

- **Planned Trajectory (Blue)**: Target position requested by G-code.
  - **Actual Feedback Position (Red)**: Encoder / motor feedback reported by controller.
  - **Position Error (Green)**: Difference  $\Delta z(t) = z_{\text{actual}}(t) - z_{\text{planned}}(t)$ .
- 

### 7.2 2. Dynamic Performance Indicators

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The telemetry panel displays:

- **Current Axis Positions**: Live coordinates  $((X, Y, Z, U, V, W))$ .
  - **Actual Speed / Velocity**: Real-time motion speed in  $(\text{mm/s})$ .
  - **Buffer State**: Controller G-code command buffer fill level (prevents buffer underruns during high-frequency motion).
  - **Trigger Event Counter**: Total number of Geiger trigger pulses issued during the session.
- 

### 7.3 3. Logging & Data Export

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TRACE automatically records execution logs for motion quality assurance (QA):

- **Save Log File**: Exports execution history to CSV format (`motion_log_YYYYMMDD_HHMMSS.csv`).
- **QA Metrics**:
  - **Root Mean Square Error (RMSE)** across the entire trajectory:  $\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left( z_{\text{actual}}(i) - z_{\text{planned}}(i) \right)^2}$
  - **Max Peak Position Error**.
  - **Trigger Jitter**: Timing variation between planned phase and actual electrical pulse.

## 8. Troubleshooting & Safety Guide

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This guide covers common operational issues, hardware error messages, recovery procedures, and safety protocols when running TRACE with the dynamic phantom.

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### 8.1 Hardware Connection Issues

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#### 1. "Duet Controller Disconnected / Timeout"

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**Symptoms:** Red status indicator, commands fail to send.

**Solutions:** - Verify Ethernet cable connection between workstation and Duet. - Test network connectivity:

```
ping 192.168.1.100
```

- Open browser and navigate to `http://192.168.1.100` to confirm Duet Web Control (DWC) is responding. - Verify IP address matches `duet_ip` in `configuration.json`.

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#### 2. "Geiger Serial Port COMx Not Found"

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**Symptoms:** Cannot toggle trigger synchronization or test pulses.

**Solutions:** - Unplug USB-to-Serial cable and re-insert. - Open Device Manager (Windows) or run `ls /dev/tty*` (Linux) to check assigned port number. - Update `geiger_port` in `configuration.json` and restart TRACE.

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### 8.2 Motion & Limit Switch Errors

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#### 3. "Axis Hit Limit Switch / Endstop Triggered"

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**Symptoms:** Emergency stop activated, movement freezes mid-trajectory.

**Solutions:** 1. Check physical platform position for mechanical obstruction or extreme travel. 2. Click **Reset Controller Alarm**. 3. Use manual jog controls in reverse direction at low speed ( $(0.1\text{ mm})$ ) to backed off switch. 4. Verify motion trajectory limits in **Planning Tab** before re-running.

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#### 4. "Buffer Underrun / Stuttering Motion"

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**Symptoms:** Phantom movement stutters or pauses intermittently.

**Solutions:** - Reduce point density / increase step interval in **Planning Tab** during G-code generation. - Close CPU-intensive background applications. - Switch connection mode from Serial to Ethernet REST API for higher throughput.

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## 8.3 Safety Protocols

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### Laboratory Safety Guidelines

- **Never place hands or tools inside the phantom motion stage while power is applied.**
- Keep the hardware Emergency Stop button within easy reach during all automated trajectory runs.
- Inspect motor belts and mechanical fasteners weekly for wear or tension loss.